

Cognitive Science — Module 07: Communication —

Study Questions

1. What does it mean that communication is "shared inference"? How does this differ from the "information transfer" metaphor? Why is the difference important?
2. Describe the coupled dynamic of communication. What does it mean that "I model you modeling me"?
3. What is Theory of Mind (ToM)? At what age does it develop in children? Why does Active Inference provide a natural framework for ToM?
4. What are first-order, second-order, and third-order Theory of Mind? Give an example of each. Why does social complexity require higher-order ToM?
5. How does language function as a shared generative model? What happens when speaker and listener have different generative models (e.g., speaking different languages)?
6. Using the "ordering coffee" example, trace all the predictions and inferences happening in both the customer's and the barista's minds during a 10-second interaction.
7. What are cultural norms in Active Inference terms? Why do norms reduce prediction errors in social interactions?
8. How does Active Inference explain echo chambers? Why do algorithms that show you "relevant" content inevitably create echo chambers?
9. How does Active Inference explain why misinformation spreads so effectively? Is the problem that people are gullible, or that misinformation fits their existing model?
10. What makes content "viral"? How does the concept of optimal prediction error level explain why some things are shared widely and others aren't?
11. How does non-verbal communication (facial expressions, body language, prosody) fit into the Active Inference framework?
12. What is neural synchronization during conversation? Is it evidence that brains literally "sync up" during communication?
13. How does Active Inference reframe disagreement? Instead of "one person is right and one is wrong," what alternative framework does it offer?
14. How might Active Inference explain sarcasm? How does the listener detect that the speaker's literal meaning differs from their intended meaning?

15. Critically evaluate: If communication is shared inference, can an AI chatbot that generates persuasive text be said to "communicate" with humans? Does it have a generative model? Is it performing theory of mind?
16. How does Active Inference explain the phenomenon of "talking past each other" — when two people are having what seems like the same conversation but are actually operating under completely different generative models? Give a concrete example and describe what would need to change for genuine understanding to occur.
17. What is the role of "repair" in communication? When a misunderstanding occurs (a large prediction error in the listener's model), what sequences of verbal and non-verbal actions do speakers use to re-align their mutual generative models? Why is conversational repair so rapid and often automatic?
18. How does written communication differ from spoken communication in Active Inference terms? What information channels are lost (prosody, facial expression, timing) and how does this change the inference problem for the reader? Why are misunderstandings more common in text messages and emails?
19. Describe how Active Inference applies to cross-cultural communication. When people from different cultural backgrounds interact, their generative models may have different priors about social norms, politeness, and meaning. How does the framework explain both successful cross-cultural communication and systematic misunderstandings?
20. How might the Active Inference framework inform the design of better social media platforms? If current algorithms optimize for engagement (maximizing prediction error / surprise), what would a platform look like that instead optimized for genuine model alignment and productive disagreement?